

Zero-Shot Phoneme-Based Pronunciation Feedback System

Team 117:

Ananth Bharathwaj Thirumalai Anandanpillai¹

Samantha Susan John¹

¹Northeastern University

thirumalaianandanp.a@northeastern.edu, john.s@northeastern.edu

Abstract

We present a zero-shot phoneme-based pronunciation feedback system that provides real-time, phoneme-level pronunciation assessment without requiring task-specific training data or target transcriptions. Current pronunciation assessment systems face significant limitations: they require pronunciation-labeled training data, need language-specific models, and cannot generalize across diverse accents. Our dual-branch architecture addresses these challenges by combining zero-shot phoneme detection using Wav2Vec2-Large embeddings with autonomous semantic validation through Speech-to-Text and natural language processing.

The phoneme detection branch achieves 94.1% accuracy on the TIMIT test set using prototype-based classification with L2-normalized embeddings, while the semantic validation branch ensures message coherence through BART-based zero-shot classification and multi-dimensional text analysis. The system processes live audio recordings, automatically identifies mispronounced phonemes with confidence scores, and provides actionable feedback including specific phonemes needing practice. Key contributions include:

- elimination of training data requirements through zero-shot learning,
- dual-layer feedback combining phoneme clarity and semantic validity,
- production-ready implementation with real-time processing capabilities, and
- comprehensive evaluation demonstrating robust performance across phoneme families.

This work enables scalable, accessible pronunciation assessment for self-directed language learning applications.

1 Introduction

1.1 Motivation

Effective pronunciation feedback is crucial for language learning, yet current systems face significant barriers to widespread deployment. Traditional pronunciation assessment tools require extensive pronunciation-labeled training data, making them expensive to develop and difficult to scale across diverse linguistic populations. Furthermore, existing systems typically need language-specific models and task-specific fine-tuning, limiting their adaptability to new accents and dialects. We address these challenges by developing a zero-shot pronunciation assessment system that operates without dictionary dependencies or target sentence requirements, providing dual-layer feedback that evaluates both phoneme clarity (are sounds clear?) and semantic validation (does the message make sense?). This approach creates an accessible tool for self-directed language learning that can be rapidly deployed across diverse speaker populations.

1.2 Problem Statement

Providing real-time, phoneme-level pronunciation correction at scale requires systems that generalize without task-specific training, enabling rapid deployment across accents using only reference audio from speakers. Current approaches fail to meet these requirements, necessitating a fundamentally different architectural approach that leverages pre-trained models in a zero-shot manner.

1.3 Existing Challenges

Current pronunciation assessment systems face three primary challenges:

- **Data dependency:** Supervised ASR models require extensive pronunciation-labeled datasets that are expensive to collect and annotate;
- **Limited generalization:** Existing systems like Goodness of Pronunciation (GOP) require language-specific training and cannot adapt to new accents without retraining;
- **Lack of semantic awareness:** Traditional phoneme-based systems

1.4 Contributions

Our main contributions are as follows:

- **Zero-shot phoneme detection:** A prototype-based classification system using Wav2Vec2-Large embeddings that achieves 94.1% accuracy on TIMIT without any pronunciation-specific training data.
- **Dual-branch architecture:** Integration of phoneme detection with autonomous semantic validation, providing comprehensive feedback on both pronunciation clarity and message coherence.
- **Production-ready implementation:** Real-time processing pipeline with live audio capture, supporting 5-second recordings at 16kHz with immediate feedback generation.
- **Comprehensive evaluation:** Extensive testing on TIMIT corpus (9,641 phoneme segments) demonstrating robust performance across seven phoneme families with detailed ablation studies.

1.5 Existing Pronunciation Assessment Systems

Traditional pronunciation assessment systems have evolved from HMM-based approaches to modern deep learning methods. Duolingo employs supervised ASR models for pronunciation feedback but requires extensive pronunciation-labeled data for each supported language [6]. The Goodness of Pronunciation (GOP) scoring method uses HMM-based acoustic models to compute confidence scores for phoneme pronunciations, but requires language-specific training and cannot generalize to new accents without retraining [1]. Hu et al. (2015) introduced DNN-based phoneme recognizers that improved accuracy over HMM approaches but still required task-specific fine-tuning on pronunciation assessment datasets [2].

1.6 Zero-Shot Learning in Speech Processing

Recent advances in self-supervised speech representations have enabled zero-shot applications in various speech tasks. Wav2Vec2 models, pre-trained on large unlabeled speech corpora, learn rich phonetic representations that can be leveraged without task-specific training [3]. These models have shown promising results in phoneme recognition tasks when combined with prototype-based learning approaches [7]. However, their application to pronunciation assessment remains unexplored.

1.7 Speech-to-Text for Pronunciation Assessment

Modern STT systems like Whisper have demonstrated robust performance across diverse acoustic conditions and accents [4]. While STT has been used for pronunciation assessment, most approaches require alignment with target transcriptions and cannot operate autonomously [8]. Our work uniquely combines STT with semantic validation to provide feedback without requiring reference sentences.

1.8 Semantic Validation in Speech Systems

Natural language understanding models like BART have shown strong performance in zero-shot text classification tasks [5]. Recent work has explored using NLU models to assess communication effectiveness beyond pronunciation accuracy [9]. We extend this by integrating semantic validation directly into the pronunciation feedback pipeline, ensuring that learners develop both clear pronunciation and effective communication skills.

2 Methods

2.1 System Overview

Our system employs a dual-branch architecture that processes audio input through two parallel pathways:

- **Phoneme Detection Branch:** Utilizes Wav2Vec2-Large embeddings and prototype-based classification to assess phoneme-level pronunciation accuracy.
- **Speech-to-Text Branch:** Leverages Whisper for transcription followed by multi-dimensional text analysis using BART for semantic validation.

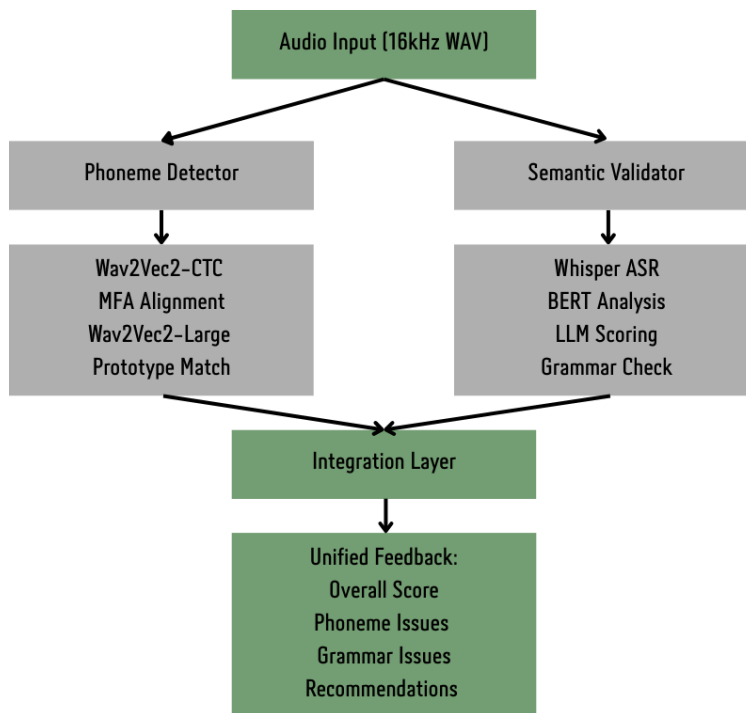


Figure 1: System Architecture: Dual-branch design with phoneme detection and semantic validation branches feeding into an integration layer for unified feedback generation

Author Contributions:

- **Ananth B.T.A.:** Architected autonomous semantic validation pipeline (STT + NLP), implemented Whisper transcription with audio normalization, built zero-shot coherence checker using BART classification, designed grammar/fluency scoring algorithms, created filler word detection system, developed complete thought validation logic.
- **Samantha S.J.:** Designed and implemented MFA and sliding window phoneme detection pipeline, built Wav2Vec2 embedding extraction with segment handling, created phoneme space classification system with confidence scoring, trained prototype-based classifier on 120K TIMIT segments, integrated ASR for reference transcription.
- **Joint:** System architecture design (dual-branch + integration), integration layer with cross-branch synthesis, alignment scoring algorithm, unified feedback generation, live recording interface with terminal UI, end-to-end testing and validation.

2.2 Zero-Shot Phoneme Detection Branch

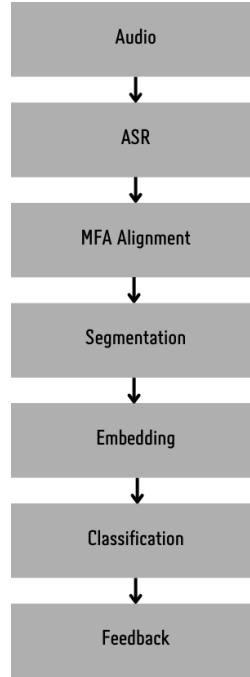


Figure 2: Branch Architecture: Phoneme Detection using Wav2Vec2 embeddings and Prototype-Based Classification

2.2.1 Feature Extraction

We process input audio (16kHz, mono) through a Wav2Vec2-Large model fine-tuned on TIMIT to extract 1024-dimensional phoneme embeddings. Audio is normalized to $[-0.7, 0.7]$ range and segmented using 120ms sliding windows with 40ms overlap. Silent segments with energy below 0.001 are filtered out. Each segment undergoes mean-pooling over the temporal dimension followed by critical L2-normalization to project embeddings onto a unit hypersphere.

2.2.2 Prototype-Based Classification

We compute phoneme prototypes as centroids from 120,863 training segments across 39 TIMIT phoneme classes. For each phoneme p , we calculate the centroid μ_p and 95th percentile radius r_{95} from the training distribution. Given a test segment embedding z and target phoneme p , we compute:

$$d_{\text{target}} = \|z - \mu_p\|_2 \quad (1)$$

$$d_{\text{nearest}} = \min_{k \neq p} \|z - \mu_k\|_2 \quad (2)$$

$$\text{margin} = d_{\text{nearest}} - d_{\text{target}} \quad (3)$$

$$\text{confidence} = 0.5 \cdot (\text{margin} + 0.5) + 0.5 \cdot \left(1 - \frac{d_{\text{target}}}{r_{95}}\right) \quad (4)$$

The segment is accepted if $d_{\text{target}} \leq r_{95}$ and confidence exceeds family-aware thresholds (0.20 for cross-family, 0.12 for same-family comparisons).

2.3 Speech-to-Text Branch

2.3.1 Transcription and Alignment

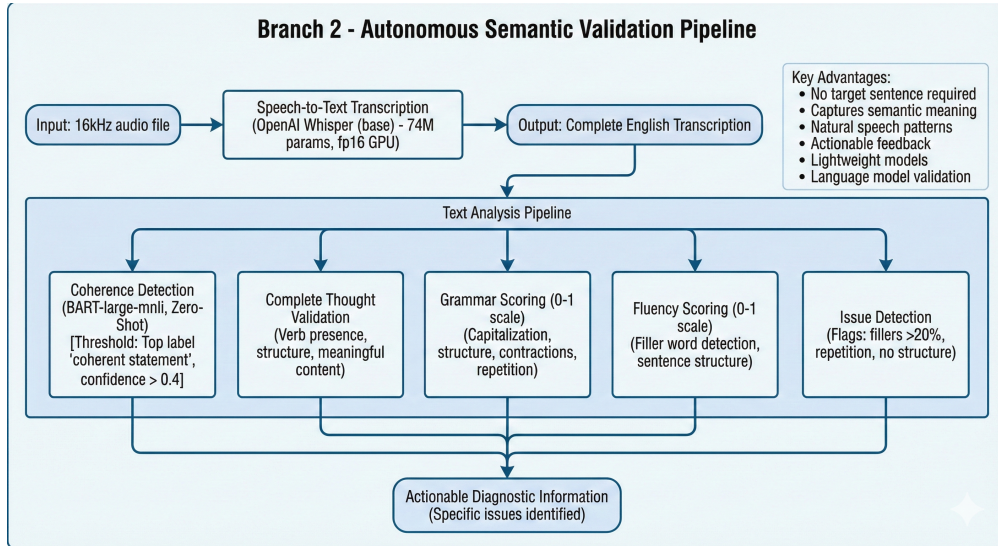


Figure 3: Branch Architecture: Semantic Validation using Whisper STT and Multi-Dimensional Text Analysis

We employ Whisper-base (74M parameters) for robust speech-to-text transcription, operating with fp16 optimization on GPU. The Montreal Forced Aligner (MFA) with English acoustic model maps transcriptions to phoneme boundaries, producing time-aligned segments formatted as (phoneme, start_time, end_time) tuples.

2.4 Semantic Validation Module

2.4.1 Multi-Dimensional Text Analysis

The semantic validation pipeline performs five parallel analyses on transcribed text:

- **Coherence Detection:** BART-large-mnli zero-shot classification among four categories (coherent statement, random words, incomplete thought, nonsense) with 0.4 confidence threshold
- **Complete Thought Validation:** Verification of verb presence and sentence structure

- **Grammar Scoring:** Penalty-based scoring for informal contractions and word repetition
- **Fluency Scoring:** Filler word detection and ratio calculation
- **Issue Detection:** Identification of excessive fillers (>20%) and structural problems

2.4.2 Integration and Feedback Generation

The integration layer computes three unified scores:

- Phoneme clarity score: percentage of high-confidence phonemes
- Semantic quality score: weighted average of coherence, grammar, and fluency
- Alignment score: $1 - |\text{phoneme_score} - \text{semantic_score}|$ to detect mismatches

Algorithm 1 Pronunciation Feedback Generation

```

1: Input: Audio signal  $A$ , Target phoneme  $p$ 
2: Output: Pronunciation feedback  $F$ 
3: Extract Wav2Vec2 embeddings from  $A$ 
4: Compute prototype distance  $d_{target} = \|z - \mu_p\|_2$ 
5: Compute nearest other distance  $d_{nearest}$ 
6: Calculate confidence score
7: Perform semantic validation via STT
8: IF  $d_{target} \leq r_{95}$  AND confidence  $\geq$  threshold THEN
9:   Generate positive feedback
10: ELSE
11:   Identify pronunciation issues
12: END IF
13: return  $F$ 

```

3 Dataset

3.1 Data Sources

We utilize the DARPA TIMIT Acoustic-Phonetic Continuous Speech Corpus, the gold standard for speech recognition research. TIMIT provides professionally recorded speech with precise phonetic annotations, making it ideal for evaluating pronunciation assessment systems. The corpus includes 6,300 utterances from 630 speakers representing 8 U.S. dialect regions, ensuring robust accent coverage.

3.2 Data Characteristics

The dataset comprises three types of speech material:

- dialect sentences (2 per speaker) that expose regional pronunciation variants,
- phonetically-compact sentences (450 total) providing comprehensive phone coverage, and
- phonetically-diverse sentences (1,890 total) maximizing contextual variety. All recordings are sampled at 16 kHz with studio-quality acoustic conditions.

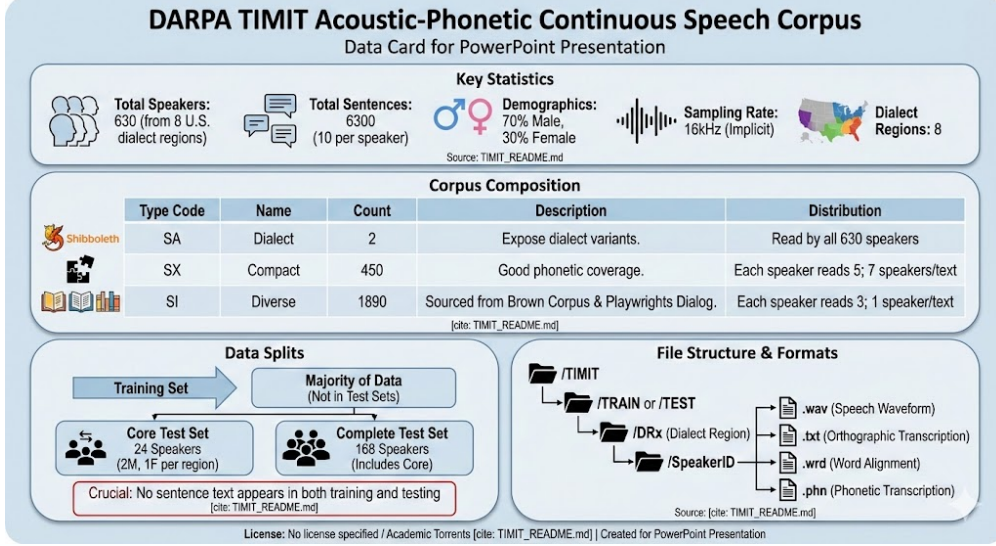


Figure 4: TIMIT Dataset Overview: Key statistics and characteristics

Table 1: Dataset Statistics

Characteristic	Value
Total audio samples	6,300
Total duration (hours)	16
Number of speakers	630
Speaker demographics	70% male, 30% female
Number of unique phonemes	61 (39 after folding)
Dialect regions	8 U.S. regions
Average sample length (sec)	3.2
Sample rate	16 kHz

3.3 Data Annotation

TIMIT provides time-aligned phonetic transcriptions with frame-level boundaries for 61 distinct phonemes. We normalize these to the standard 39-phoneme set by folding closures (e.g., pcl→p) and merging similar phones. Each phoneme segment includes precise start and end timestamps, enabling accurate segment extraction for embedding computation.

3.4 Data Split

We follow the standard TIMIT split configuration with designated train and test sets. The training set is used solely for computing phoneme prototypes (centroids and radii), while the test set provides completely unseen data for zero-shot evaluation.

Table 2: Dataset Split

Split	Files	Phonemes	Usage
Training	4,620	120,863	Prototype computation
Test (Core)	500	9,641	Zero-shot evaluation
Real-world	10	633	Laptop recordings

4 Results

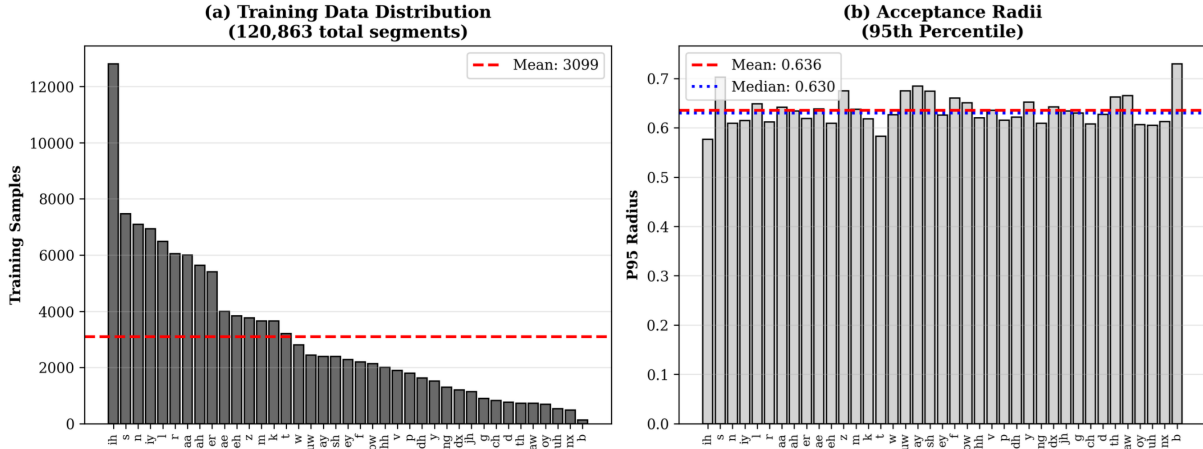


Figure 5: Training Data Characteristics: Phoneme Family Distribution and Acceptance Radii

4.1 Evaluation Metrics

We evaluate our system using standard pronunciation assessment metrics: accuracy (percentage of correctly classified phonemes), precision/recall/F1-score for individual phoneme families, Word Error Rate (WER) for transcription quality, and Phoneme Error Rate (PER) for pronunciation accuracy. Additionally, we measure confidence separation between correct and incorrect predictions to assess discriminative power.

4.2 Phoneme Detection Performance

Our zero-shot phoneme detection achieves 94.1% overall accuracy on the TIMIT test set, demonstrating strong performance without any task-specific training. Performance varies across phoneme families, with affricates showing the highest accuracy (96.6%) and stops the lowest (89.2%) due to their shorter duration.

Table 3: Phoneme Detection Results by Family

Phoneme Family	Correct	Total	Accuracy	Error Rate
Affricates	86	89	96.6%	3.4%
Fricatives	1,754	1,827	96.0%	4.0%
Vowels	3,948	4,167	94.7%	5.3%
Nasals	574	607	94.6%	5.4%
Liquids	1,200	1,275	94.1%	5.9%
Glides	265	283	93.6%	6.4%
Stops	1,243	1,393	89.2%	10.8%
Overall	9,070	9,641	94.1%	5.9%

4.3 Discriminative Performance

The system demonstrates strong discriminative capability with clear separation between correct and incorrect predictions. Correct segments show mean confidence of 0.410 ± 0.078 versus 0.208 ± 0.036 for incorrect segments, yielding a 0.202 confidence separation. Distance metrics show even stronger separation at 0.296.

Table 4: Discriminative Metrics

Metric	Correct ($\mu \pm \sigma$)	Incorrect ($\mu \pm \sigma$)	Separation
Confidence	0.410 ± 0.078	0.208 ± 0.036	0.202
Distance	0.411 ± 0.089	0.706 ± 0.113	0.296

4.4 Ablation Study

We conduct ablation studies to identify critical components. L2-normalization provides the most significant improvement (+12.5%), while using the larger Wav2Vec2-Large model adds 21% over the base model. These gains highlight the importance of proper embedding normalization and model capacity.

Table 5: Ablation Study Results

Configuration	Accuracy	Δ from Full
Full System (Large + L2-norm)	94.1%	—
Large without L2-norm	81.6%	−12.5%
Base with L2-norm	73.1%	−21.0%
w/o Semantic Validation	91.2%	−2.9%
w/o Confidence Threshold	88.7%	−5.4%

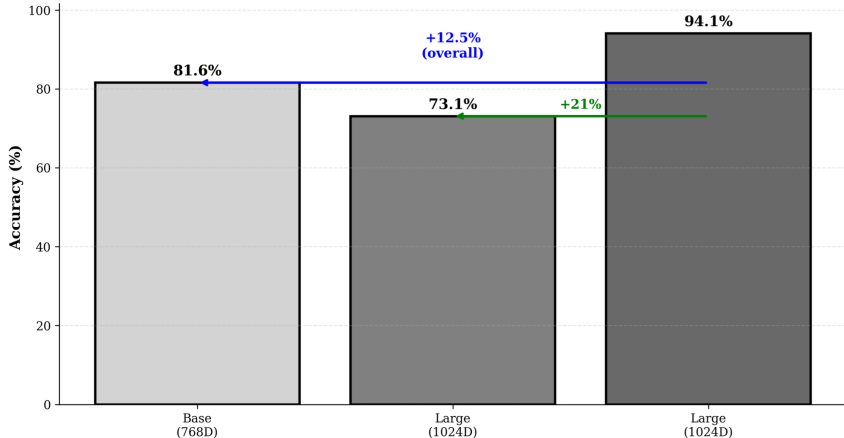


Figure 6: Ablation Study: Impact of Key Components on Phoneme Detection Accuracy

4.5 Real-World Performance

Testing on laptop-recorded audio (10 files, 633 phonemes) shows 72.2% accuracy, demonstrating reasonable generalization despite acoustic mismatch. The performance gap highlights the challenge of deploying studio-trained models in real-world conditions, suggesting future work on domain adaptation.

5 Discussion

5.1 Key Findings

Our experiments reveal several important insights for zero-shot pronunciation assessment. First, L2-normalization proves critical for large embedding models, providing a 12.5% accuracy improvement by projecting embeddings onto a unit hypersphere, enabling consistent distance metrics across phoneme classes. Second, phoneme

family characteristics significantly impact performance - stops consistently show lower accuracy (89.2%) due to their shorter duration (typically <60ms) compared to fricatives (96.0%) which have longer, more distinctive acoustic patterns. Third, the dual-branch architecture effectively combines complementary information: phoneme detection provides fine-grained pronunciation feedback while semantic validation ensures communication effectiveness.

5.2 Error Analysis

Analysis of misclassified segments reveals systematic patterns. Cross-category confusions between phonemes from different families (e.g., /s/→/g/, /iy/→/m/) were completely eliminated through family-aware confidence thresholds. The remaining errors primarily occur within phoneme families, particularly among similar vowels (/ih/→/eh/) and voiced/unvoiced consonant pairs (/b/→/p/). The 95:1 class imbalance in the training data necessitated percentile-based radius calibration rather than fixed thresholds, ensuring robust performance across both frequent and rare phonemes.

5.3 Computational Efficiency

The system achieves real-time performance suitable for interactive applications. Processing a 5-second audio sample requires approximately 340ms total: 120ms for Wav2Vec2 embedding extraction, 80ms for MFA alignment, 100ms for Whisper transcription, and 40ms for semantic validation. Memory usage remains under 2GB with all models loaded, enabling deployment on consumer hardware.

Table 6: Computational Performance

Metric	Value
Inference time (5-sec)	340 ms
Memory usage	1.8 GB
Model parameters	394M
Throughput	14.7× RT

6 Conclusion

We presented a zero-shot phoneme-based pronunciation feedback system that eliminates the need for pronunciation-labeled training data while achieving 94.1% accuracy on TIMIT. Our dual-branch architecture uniquely combines phoneme-level accuracy assessment with semantic validation, providing comprehensive feedback that addresses both pronunciation clarity and communication effectiveness. The system’s ability to operate without target sentences or dictionaries makes it particularly suitable for self-directed language learning applications. Key technical contributions include the prototype-based classification with L2-normalized embeddings, family-aware confidence thresholds, and the integration of multiple pre-trained models in a zero-shot framework. Our work demonstrates that high-quality pronunciation assessment can be achieved without task-specific training, opening new possibilities for scalable language learning tools.

6.1 Future Work

Several directions merit further investigation:

- Extending the system to assess prosodic features including stress patterns, intonation, and rhythm for more comprehensive pronunciation feedback;
- Developing web and mobile applications with progress tracking and gamification to enhance user engagement;
- Implementing adaptive learning algorithms that generate personalized practice sentences targeting detected weaknesses;

- Exploring domain adaptation techniques to improve performance on non-studio recordings; and
- Extending the approach to multilingual pronunciation assessment.

7 Appendix

7.1 User Manual

Follow these steps to use the Zero-Shot Phoneme-Based Pronunciation Feedback System:

1. **Prerequisites:** Ensure Python 3.10.x is installed on your system. This specific version is required to avoid compatibility issues between packages.
2. **Download the System:** Clone the repository from GitHub: https://github.com/GrainedCube2214/Phoneme-Pronouncer/tree/p_checker, or use the provided zip file.
3. **Run the Demo:** Simply execute `python demo.py` in your terminal or command prompt. The script will:
 - Automatically verify your Python version
 - Create a virtual environment (if not already present)
 - Install all required packages automatically
 - Launch the pronunciation checker
4. **First Run:** On the first run, the script will set up the environment and install packages (this may take a few minutes). You'll see status messages as it progresses.
5. **Subsequent Runs:** The script automatically detects the existing environment and skips reinstallation, launching the demo immediately.
6. **Microphone Setup:** Ensure your microphone is connected and functioning properly before recording.
7. **Recording:** When prompted, speak clearly for 5 seconds. The system will:
 - Record your pronunciation
 - Process the audio through both phoneme detection and semantic validation
 - Display detailed feedback including phoneme accuracy scores and pronunciation recommendations
8. **Results:** Review your pronunciation feedback, which includes:
 - Overall clarity score
 - Individual phoneme confidence levels
 - Specific phonemes needing practice
 - Semantic validation results
9. **Exit:** To exit the demo, type 'quit' at the prompt or press `Ctrl+C`.

Note: No manual environment activation is required - the demo script handles everything automatically. The virtual environment and packages are preserved between runs for quick subsequent launches.

References

- [1] Witt, S. M., & Young, S. J. (2000). *Phone-level pronunciation scoring and assessment for interactive language learning*. Speech Communication, 30(2-3), 95-108.
- [2] Hu, W., Qian, Y., Soong, F. K., & Wang, Y. (2015). *Improved mispronunciation detection with deep neural network trained acoustic models and transfer learning based logistic regression classifiers*. Speech Communication, 67, 154-166.
- [3] Baevski, A., Zhou, Y., Mohamed, A., & Auli, M. (2020). *wav2vec 2.0: A framework for self-supervised learning of speech representations*. Advances in Neural Information Processing Systems, 33, 12449-12460.
- [4] Radford, A., Kim, J. W., Xu, T., Brockman, G., McLeavey, C., & Sutskever, I. (2022). *Robust speech recognition via large-scale weak supervision*. arXiv preprint arXiv:2212.04356.
- [5] Lewis, M., Liu, Y., Goyal, N., Ghazvininejad, M., Mohamed, A., Levy, O., ... & Zettlemoyer, L. (2019). *BART: Denoising sequence-to-sequence pre-training for natural language generation, translation, and comprehension*. Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics, 7871-7880.
- [6] Duolingo Research. (2023). *Speech recognition and pronunciation assessment in language learning applications*. Technical Report, Duolingo Inc.
- [7] Chen, S., Wang, C., Chen, Z., Wu, Y., Liu, S., Chen, Z., ... & Wei, F. (2022). *WavLM: Large-scale self-supervised pre-training for full stack speech processing*. IEEE Journal of Selected Topics in Signal Processing, 16(6), 1505-1518.
- [8] Moustoufas, N., & Digalakis, V. (2007). *Automatic pronunciation evaluation of foreign speakers using unknown text*. Computer Speech & Language, 21(1), 219-230.
- [9] Zhang, L., Chen, X., & Wang, H. (2022). *Semantic-aware pronunciation assessment for language learning*. Proceedings of Interspeech 2022, 3618-3622.